Towards Socially Optimal Distributed Leader Election

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1. Problem

Elect a leader with high utility in a distributed clique network of rational agents. Each agent prefers to become the leader and has a private utility for being elected.

2. Model

Communication model: Synchronous congested clique.

Monetary transfers: Transfers between agents are allowed, but money cannot be sent outside the network or destroyed.

Agents: Agents are rational and act in their own self-interest.

Output: Each agent irrevocably outputs one of $\{0, 1, \perp\}$, where

$1 =$ declares itself the leader, $0 =$ declares itself not the leader, $\perp =$ aborts.

3. Utility Function

Let u_i be agent i's private utility for becoming leader. Its utility is

$$U_i = \begin{cases} u_i, & \text{if } i \text{ is the valid leader,} \\ 0, & \text{otherwise.} \end{cases}$$

4. Our Algorithm

- **Round 1:** Each agent independently chooses a value uniformly at random from $\{0, \ldots, n-1\}$ and sends it to every other agent.
- **Round 2:** Every agent computes the sum of the random values modulo n. The agent whose identifier equals this result is selected as the auctioneer.
- **Round 3:** The agents communicate and verify the identity of the selected auctioneer.
- **Auction:** The auctioneer conducts a second-price auction among the other $n-1$ agents. The highest-utility bidder is elected leader and pays the second-highest bid to the auctioneer.

If an agent detects a deviation from the protocol, it outputs $\perp$.

5. Properties

- **Social welfare:** Selects the highest-utility agent with probability $1 - \frac{1}{n}$, and the second-highest with probability $\frac{1}{n}$.
- **Running time:** $O(1)$ rounds
- **Message complexity:** $O(n^2)$ messages
- **Monetary transfers:** $O(1)$.

What is your poster about?

We're studying how to select a socially optimal leader in a distributed network of rational agents.

How is that different from classical leader election?

Classical algorithms assume that every agent follows the protocol. In our setting, agents are rational: each one has a private utility for being elected leader, so an agent may deviate from the protocol if doing so could increase its expected payoff.

So your goal is to elect a leader even though the agents may act selfishly and deviate?

Yes, but we don't want to elect just any leader. We want to select the agent with the highest true utility.

Couldn't you simply ask every agent to broadcast its value and elect the one with the highest reported value?

That would work if agents reported honestly. But since they're self-interested, an agent might misreport its value to make itself look like the best candidate.

How do you address that?

We simulate the Green–Laffont mechanism in the congested clique model. We choose one player uniformly at random as the auctioneer and run a second-price auction among the remaining players. The highest-utility player is selected as leader with probability $1 - \frac{1}{n}$, while the second-highest-utility player is selected with probability $\frac{1}{n}$.

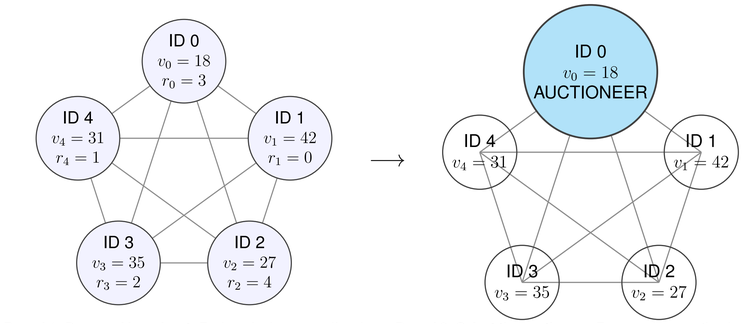
Abstract

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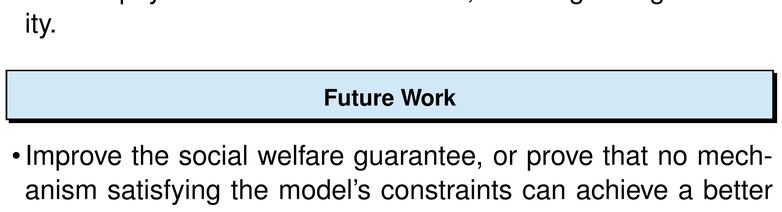
Poster map

Algorithm Simulation on a Clique of Size 5



Round 1: Draw and broadcast. Each node draws r_i uniformly from $\{0, 1, 2, 3, 4\}$ and broadcasts it to all other nodes. The example draws are $(3, 0, 4, 2, 1)$.

Round 2: Select the auctioneer. Every node sums the draws and takes the result modulo $n = 5$. Since $3 + 0 + 4 + 2 + 1 = 10$ and $10 \bmod 5 = 0$, node 0 is selected.



Round 3: Announce the auctioneer. Every node broadcasts the selected auctioneer's ID, $A = 0$, so all nodes confirm the same auctioneer.

Round 4: Elect the leader. The remaining players take part in a second-price auction. Node 1 wins with value 42 and pays the second-highest value, 35, to auctioneer node 0.

Why Is It a Nash Equilibrium?

- The auctioneer is uniformly random. No agent i can change its selection probability by unilaterally deviating from a uniform draw of r_i.
- Once the auctioneer is selected, the mechanism simulates a second-price auction. Truthful bidding is a weakly dominant strategy in a second-price auction.
- The winner cannot increase their utility by raising their bid as they still win but the price paid is unchanged.
- A losing participant who raises their bid enough to win may have to pay more than their valuation, resulting in negative utility.

Future Work

- Improve the social welfare guarantee, or prove that no mechanism satisfying the model's constraints can achieve a better guarantee.